Natural Selection Experiment

Aim: To demonstrate the process of natural selection

Materials: 3 pieces of furry fabric in 3 different colours, 3 sets of 60 pom poms (30 each of 2 colours matching the fabric), stopwatches, foam cups, plastic spoons

What to do

1. Spread the pom poms out on your piece of fabric.
2. The hunter will be given a foam cup and a spoon. They must eat the prey by picking it up with the spoon and putting it in the cup.
3. The hunter has 20 seconds to 'eat' as many animals as they can.
4. Count the number of animals of each type eaten and record it in the table below
5. Repeat the experiment twice more with the same fabric.
6. Change to a different fabric and repeat the process.

Results

Fur Colour:

	Trial 1	Trial 2	Trial 3	Average
Number of black animals at start				
Number of black animals eaten				
Number of black animals left				
Number of white animals at start				
Number of white animals eaten				
Number of white animals left				

Fur Colour:

	Trial 1	Trial 2	Trial 3	Average
Number of black animals at start				
Number of black animals eaten				
Number of black animals left				
Number of white animals at start				
Number of white animals eaten				
Number of white animals left				

Fur Colour:

	Trial 1	Trial 2	Trial 3	Average
Number of black animals at start				
Number of black animals eaten				
Number of black animals left				
Number of white animals at start				
Number of white animals eaten				
Number of white animals left				

Conclusions

What do you notice about how many animals are eaten and the colour of the background fabric?

How can this be related to real animals and the adaptations they have?

If your pom poms were real animals in the wild, they would reproduce. Assuming each animal produces offspring with the same adaptations, in this case colour, what do you think would happen to the population over successive generations?